

# JOSEPH ALAN B. VERGARA

• Phase II-Doña Maria Subd, Tubod, Lanao Del Norte • <https://www.joalvergs.tech/> • (+63) 947-589-2995

## EDUCATION

### BS Computer Applications Major in Embedded Systems

4th Year Student at MSU- Iligan Institute of Technology | **CGPA: 1.96269**

**August 2023 - Present**

Iligan City, Philippines

### Senior High School | ABM Strand

**2021-2023**

MSU-IIT Integrated Developmental School | **Academic Honors**

A. Bonifacio Ave, Brgy. Tibanga, Lanao Del Norte

## SKILLS

|                                             |                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming Language:</b>                | C, C++, Python, JavaScript, TypeScript, Java, PLC (Ladder Logic), Assembly, Bash/Shell Scripting, CSS                                                                                                                                                                                                                                                              |
| <b>Operating Systems &amp; CLI:</b>         | Linux (Arch Linux, CachyOS), Unix Shell & CLI Configuration (Zsh, Bash, Dotfiles), RT-Thread RTOS                                                                                                                                                                                                                                                                  |
| <b>Networking &amp; Web Infrastructure:</b> | DNS Management (A, CNAME, TXT records, Nameservers), Custom Domain Configuration, SSL/TLS, Vercel, Railway, Hostinger, Cisco Packet Tracer                                                                                                                                                                                                                         |
| <b>Software &amp; Frameworks:</b>           | Visual Studio Code, Git, GitHub, Node.js, React.js, Express.js, Pictoblox, CodeBlocks, Microchip (ATMEL) Studio, Arduino IDE, Google Colab, Edge Impulse, TensorFlow, MATLAB, MARIE.js, Vivado, RT-Thread Studio, SquareLine Studio, MongoDB, Supabase                                                                                                             |
| <b>Hardware:</b>                            | Microcontrollers (i.e., 8051, ATmega328P, ATmega2560, ESP8266, ESP32), Field Programmable Gate Arrays (FPGAs), Basic and Advanced Electronics, Test Equipment (i.e., Multimeter, Oscilloscope), Prototyping (i.e., PCB layout, design, prototyping, simulation, soldering), Computer & Laptop Hardware Repair (Teardown, Component Diagnostics, Parts Replacement) |
| <b>Electronics:</b>                         | Applied Math & Calculus, Principles of Electric Circuits, Electronics Fundamentals, Digital Design, Computer Architecture                                                                                                                                                                                                                                          |
| <b>Soft Skills:</b>                         | <b>Vibe - coding</b> Cross-Functional Team Collaboration Abilities, Organizational & Time Management Skills, Being Self-Driven and Independent, Adaptability & Patience                                                                                                                                                                                            |
| <b>Spoken Language:</b>                     | English, Filipino, Cebuano                                                                                                                                                                                                                                                                                                                                         |

## PROJECTS

### Microcontroller-Based Automated Feedback and Scoring System (In Progress)

Co-Author & Full-Stack / Embedded Developer | PhD Dissertation Research Collaboration

- Designed a specialized full-stack Learning Management System (React.js, Node.js, Supabase/PostgreSQL) to capture live microcontroller telemetry and automate laboratory scoring logic.
- Reduced evaluation latency and eliminated manual record-keeping errors by engineering deterministic evaluation pipelines paired with interactive instructor dashboards for instant feedback.
- Benchmarked standard web server deployments against an edge-hosted architecture on the ESP32-P4 microcontroller to evaluate low-power, localized server reliability. | *Live Demo:* [www.miow-lms.ids-ter.online](http://www.miow-lms.ids-ter.online) | *GitHub:* <https://github.com/MIOW-CODES/lms>

### Mobile Application Development for Computer Vision AI Development in Detecting Microplastics (In Progress)

Co-Author & Mobile/Edge AI Developer | Undergraduate Research Collaboration

- Deployed an optimized YOLOv8n edge AI model into a cross-platform mobile app for zero-latency, on-device particle quantification.
- Configured Android SDK build environments to generate release APKs and established deployment bundles for iOS targets.
- Enhanced UI/UX flow to handle camera lifecycles, live bounding-box rendering, and research-grade inference visualization. | *Documentation:* [https://drive.google.com/drive/folders/1GOxtqUJavVIL\\_n6gHiRzwvTUOWCLO-tB?usp=sharing](https://drive.google.com/drive/folders/1GOxtqUJavVIL_n6gHiRzwvTUOWCLO-tB?usp=sharing) | *GitHub:* <https://github.com/MP-DETECT-CODE>

## Barangay Connect Web Application

Full-Stack Developer | Web Development, GitHub

- Built a full-stack portal streamlining local public service requests, document generation, and real-time community reporting. | GitHub: <https://github.com/Joal0816/Barangay-Connect>

## Edge AI Glasses Object Detection

Edge AI Developer | Roboflow, Edge Impulse

- Trained, quantized, and validated a low-latency edge vision model for wearable hardware via Edge Impulse and custom Roboflow datasets. | Project: <https://studio.edgeimpulse.com/studio/848525>

## Industrial Oven Simulation & Control System

PLC & Automation Developer | CODESYS V3, PLC, HMI

- Engineered virtual PLC ladder logic and an interactive HMI with automated heating logic, safety interlocks, and threshold alarms. | Documentation: <https://sites.google.com/g.msuiit.edu.ph/hassanvergarafinalproject?usp=sharing>

## Snake OS: Embedded Interactive HMI Game

Embedded Developer | Renesas RA6M3, RT-Thread OS, LVGL

- Engineered an arcade game engine utilizing RT-Thread RTOS and LVGL to deliver fluid real-time graphical rendering on bare metal.
- Implemented autonomous AI autopilot navigation, non-volatile high-score persistence, and dynamic multi-level obstacle logic. | Documentation: <https://www.hackster.io/josephalanvergara/snake-os-interactive-hmi-game-on-rt-thread-f8b988>

## Line Following Robot

Firmware Developer | Arduino Uno, AVR Assembly

- Hand-tuned AVR Assembly firmware and dual PWM hardware timers (Timer0/2) for a 5-channel IR array, achieving a 100% course completion rate. | Documentation: <https://docs.google.com/document/d/1k9SudemZJafofm2p3hludQYK7nMUtvUpkMm6N-rJvY/edit?usp=sharing>

**Adjustable DC Power Supply:** Engineered a power supply unit with variable output to 12V and short-circuit protection.

## TRAININGS & WORKSHOPS (CERTIFICATIONS)

### AI Career Readiness Training

ICT 3D, CCS Building, MSU-IIT, Iligan City

August 20, 2026

### Quantum Circuits as Pictures: An Introduction to ZX Calculus

Zoom Meeting, Webinar

March 20, 2026

### Permaculture Webinar “From Code to Crops”

Zoom Meeting, Webinar

December 22, 2025

### my.ComApps Technology Symposium and Exhibit on RT-Thread Based Technology

4th floor, CCS Building, MSU-IIT, Iligan City

November 24-25, 2025

### Seize Your Opportunity Internships and OJTs Workshop!

ICT 3D, CCS Building, MSU-IIT, Iligan City

November 24, 2025

### Software Freedom Day 2023

CCS Building, MSU-IIT, Iligan City

September 16, 2023

## AWARDS & LEADERSHIPS

### IEEE Sumpai Hackathon 2026

1st floor, CCS Building, ICT1D, MSU-IIT, Iligan City

September 14, 2026

**3rd placer and Top 5 Finalist**; developed AgapAI, an AI-powered emergency response platform that transforms natural voice reports into structured, triage-ready incident data for dispatcher decision support.

### TechShowcase 2025 Innovation in Application Development and Emerging Technologies

PRISM Minitheater, MSU-IIT, Iligan City

December 17, 2025

**2nd placer** for best innovation, **3rd placer** for best pitch; developed Barangay Connect, a full-stack React + Supabase platform enabling residents to report incidents, request documents, and vote in polls, streamlining barangay governance.

### College of Computer Studies Executive Council - Sports & Development Affairs - Undersecretary

MSU- Iligan Institute of Technology, Iligan City, Philippines

August 2025 – May 2026

### Titans Esports - Mobile Legends BangBang Team Captain

MSU- Iligan Institute of Technology, Iligan City, Philippines

2021-2023